Zippers: an informal note on the mathematics behind the programs

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This is an informal note, *not for publication*. It describes some elements of the mathematical theory of zippers, as the programs of the `zippers` suite illustrate them. Theorems 4.1 and 6.1 are work in progress: what is given for them is a sketch of a proof, not a proof.

1 Zippers

Let M be a closed manifold with word-hyperbolic fundamental group G, and let $\varphi\colon G \to \mathbb{Z}$ be a surjective homomorphism, for instance the one induced by a map $M \to S^1$. Write $K = \ker \varphi$. The universal cover $\widetilde{M}$ is δ-hyperbolic, and G acts on it and on its Gromov boundary $\partial_\infty G$. The map $M \to S^1$ lifts to a function $f\colon \widetilde{M} \to \mathbb{R}$ with $f(gx) = f(x) + \varphi(g)$: a height.

A *staircase* is a ray in $\widetilde{M}$ that goes to infinity at a linear speed, and along which the height f grows at a linear rate. (Precisely: a quasigeodesic ray along which f increases linearly.) Every staircase converges to a point of $\partial_\infty G$. Following Calegari and Loukidou [1]:

$$Z^+ = \{\text{the endpoints of staircases}\}, \qquad Z^- = \{\text{the endpoints of staircases for } -f\}.$$

Theorem 1.1 (Calegari–Loukidou [1])**.** *If K is finitely generated, then Z^+ and Z^- are nonempty, disjoint, path-connected and invariant under G; in particular both are dense in $\partial_\infty G$.*

They are disjoint for a simple reason: two rays with the same endpoint stay a bounded distance apart, so they cannot have heights going to $+\infty$ on one and $-\infty$ on the other.

Remark 1.2. Theorem 1.1 holds more generally whenever $\varphi\colon G \to \mathbb{Z}$ is a quasimorphism whose coarse level sets are coarsely connected. In this note we consider only homomorphisms, for simplicity.

Remark 1.3. If M has cusps, so that G is relatively hyperbolic, the sets Z^+ and Z^- can still be constructed, but they are no longer disjoint.

2 How the programs draw them: base arcs from a presentation

The input. The programs start from a *presentation* of the situation:

- a finite set $S \subset K$ that generates K and contains the inverse of each of its elements (the *letters*);
- an element $t \in G$ with $\varphi(t) > 0$;

- for each letter s, its *roof*: an expression

$$t^{-1}st = x_1 x_2 \cdots x_k \qquad (x_i \in S).$$

  It exists because K is normal: $t^{-1}st$ lies in K, so it is a word in the letters.

The base arcs. Let V be the attracting fixed point of t on $\partial_\infty G$. For each letter s there is an arc α_s in $\partial_\infty G$ from V to sV, determined by the self-similarity

$$\alpha_s \;=\; t \cdot \big(\alpha_{x_1} \cup x_1\alpha_{x_2} \cup x_1x_2\alpha_{x_3} \cup \cdots \cup x_1 \cdots x_{k-1}\alpha_{x_k}\big).$$

The right-hand side is a path: $x_1 \cdots x_{i-1}\alpha_{x_i}$ runs from $x_1 \cdots x_{i-1}V$ to $x_1 \cdots x_iV$, and applying t (which fixes V) gives a path from V to $tx_1 \cdots x_kV = stV = sV$. The arcs α_s are the *base arcs*. Every point of a base arc is the endpoint of a staircase, so the union of the translates $g\alpha_s$, $g \in G$, is a path-connected, G-invariant subset of Z^+. The same construction with t^{-1}, its attracting fixed point, and the *down roofs* $tst^{-1} = y_1 \cdots y_l$ gives the base arcs of Z^-.

Remark 2.1. The union of the translates of the base arcs is contained in Z^+. It is probably equal to Z^+, but we have not thought about this carefully.

What the programs compute. A translate $h\alpha_s$ (a *branch*) is replaced by the k branches on the right-hand side, $ht\alpha_{x_1}, htx_1\alpha_{x_2}, \ldots$, until the chord joining the two ends of each branch is shorter than the accuracy (the option `-a`); such a branch is drawn as a segment. All group elements are kept exactly, and a branch reached twice is drawn once. The picture is the union of the translates $g\alpha_s$ for the elements g of a ball of G (those with $d(o, go) \le R$, the option `-R`), nearest first. With `-R 0` it is the base arcs alone.

3 Roofs

Arrange the elements of G by their height φ. The letter s is a step from 1 to s at height 0. Its roof is a path one story up, from t to st, through the points

$$t, \quad tx_1, \quad tx_1x_2, \quad \ldots, \quad tx_1 \cdots x_k = st,$$

all at height $\varphi(t)$. Each step of the roof is again a letter, so it has a roof of its own, one story higher, and so on. These nested roofs are the *blocks* of [1]; a staircase climbs through them, and the base arc α_s is the set of the endpoints of the staircases that start on the step from 1 to s.

The presentation is not unique: S, t and the roofs can all be chosen in many ways. Different choices give different base arcs, but Z^+ and Z^- depend only on G and φ. The choice matters for the pictures:

- the kernel search of the suite (`zippers search`) finds S, t and the roofs, and checks every roof exactly;
- it can choose short roofs, and roofs that make the two orientations of an arc coincide; with them, far less of the picture is drawn twice (the overdraw of Z^- in S^4 falls from 79% to 5%; `docs/developer_guide.md`).

4 Coarse connectivity of the level sets, and connectivity at infinity

A *level set* is $L_c = f^{-1}(c) \subset \widetilde{M}$. It is *coarsely n-connected* if there is a constant D such that every map of a sphere S^i, $i \leq n$, into L_c extends to a map of the ball B^{i+1} into the D-neighbourhood of L_c. For $n = 0$ this says that L_c is coarsely connected, which holds exactly when K is finitely generated; in general, the level sets are coarsely n-connected exactly when K is of type F_{n+1} (Gromov [4], where coarse higher connectivity is introduced).

Theorem 4.1 (Calegari, work in progress)**.** *If the level sets of f are coarsely n-connected, then Z^+ and Z^- are n-connected: every map of a sphere S^i, $i \leq n$, into $Z^\pm$ extends to a map of the ball B^{i+1} into $Z^\pm$.*

The key to the first step of the proof is a lemma about the geodesics that run from a point of Z^- to the points of a compact subset of Z^+, in its weak topology. Fix a basepoint o. For $K \geq 1$ let $Z^+(K)$ be the set of the endpoints of the geodesic rays that start within distance K of o and along which the height tends to $+\infty$ and is a K-quasi-isometry to $\mathbb{R}$; define $Z^-(K)$ in the same way with $-f$. Then Z^+ is the increasing union of the sets $Z^+(K)$, and it has the *weak topology*: a set is open when its intersection with each $Z^+(K)$ is open (in the topology of $\partial_\infty G$). The sets $Z^+(K)$ are not invariant under G, but Z^+ is, and G acts on it by homeomorphisms of the weak topology. A compact subset of Z^+, such as the image of a sphere or of a null-homotopy of one, lies in a single $Z^+(K)$.

Lemma 4.2. *Let X be a compact subset of Z^+, and let p be a point of Z^-. Let $\Gamma(X)$ be the family of the geodesics from p to the points of X. Then the height f is a quasi-isometry to $\mathbb{R}$ along every geodesic γ in $\Gamma(X)$, with constants that do not depend on γ.*

Proof. Since X is compact, it lies in a single $Z^+(K)$, and p lies in some $Z^-(K')$. For $x \in X$ let r_x be a ray from near o to x along which the height is a K-quasi-isometry, and r^- a ray from near o to p along which $-f$ is a K'-quasi-isometry.

1. The union $r^- \cup r_x$ is *expanding*: any two of its points y, y' satisfy $|f(y) - f(y')| \geq d(y, y')/A - B$, with A, B depending only on K, K' and the Lipschitz constant of f. On each ray this is the quasi-isometry. For $y \in r^-$ and $y' \in r_x$ it follows from $f(y) \leq f(o) - d(o, y)/A + B$, $f(y') \geq f(o) + d(o, y')/A - B$ and the triangle inequality.

2. The geodesic γ from p to x stays within a bounded distance of $r^- \cup r_x$, since ideal polygons in a hyperbolic space are thin and the two rays start near o.

3. Being expanding passes to bounded neighbourhoods, since f is Lipschitz. So $|f(\gamma(u)) - f(\gamma(u'))| \geq |u - u'|/A' - B'$, and $f \circ \gamma$ runs from $-\infty$ to $+\infty$. A continuous function on $\mathbb{R}$ with these two properties increases coarsely: it is a quasi-isometry, with constants that do not depend on x. □

Sketch of a proof of Theorem 4.1. This is a sketch only, and the details have not been written out.

1. Let $S^i \to Z^+$ be a map, with image X, and fix a point p of Z^-. By Lemma 4.2, along each geodesic from p to a point of X the height goes from $-\infty$ to $+\infty$ at a uniformly linear rate, so the upper half of each is a staircase, with uniform constants. These geodesics vary continuously with their endpoints, and each crosses a given level set in a set of bounded size: the crossings form a coarse sphere in that level set.

2. Since the level set is coarsely n-connected, that coarse sphere bounds a coarse ball at the same level.

3. Push the ball up, story by story: each small cell at one level is spanned by a cell one story higher, as each step is spanned by its roof. This is the higher-dimensional form of the block filling that proves Theorem 1.1 (where $i = 0$, and a path of steps is filled by its roofs).

4. The pushed-up cells form a ball's worth of staircases, and their endpoints give a ball in Z^+ bounding the sphere.

The work lies in step 3: keeping the constants uniform, so that the pushed-up family is still a family of staircases, and the endpoint map is continuous.

5 Examples

5.1 Hyperbolic 3-manifolds that fiber over the circle: zippers on S^2

The figure-eight knot complement fibers over the circle with fiber a once-punctured torus, whose fundamental group is the free group $F = \langle a, b \rangle$. The monodromy is the automorphism

$$\varphi(a) = ab, \qquad \varphi(b) = bab,$$

the product of the two Dehn twists $R\colon (a, b) \mapsto (a, ba)$ and $L\colon (a, b) \mapsto (ab, b)$, written RL (applied from left to right). The group is $G = \langle a, b, t \mid t^{-1}gt = \varphi(g) \text{ for } g \in F \rangle$, the height of an element is its exponent of t, and $K = F$. So the roofs are just the monodromy:

$$t^{-1}at = ab, \qquad t^{-1}bt = bab,$$

and the two base arcs of Z^+ satisfy

$$\alpha_a = t\,(\alpha_a \cup a\,\alpha_b), \qquad \alpha_b = t\,(\alpha_b \cup b\,\alpha_a \cup ba\,\alpha_b).$$

- `zippers draw fig8 -n inf` draws them for the knot complement, which is cusped (of finite volume, but not closed).

- `zippers draw fig8 -n N` draws them for the closed orbifold obtained by replacing the puncture with a cone point of order N. It is hyperbolic for every $N \geq 2$: the fiber is then a hyperbolic 2-orbifold, a torus with one cone point, and the monodromy is pseudo-Anosov. The program finds its holonomy by following the fixed point of the trace map from the cusp.

- Any positive word in R and L with both letters gives another example (`zippers draw RRL`).

- `-C` also draws the *CaTherine wheel* [2].

5.2 The examples of Llosa Isenrich and Py: zippers on S^3

Llosa Isenrich and Py [6] construct compact complex hyperbolic manifolds $X = G\backslash B^m$ of complex dimension m, with a map to the circle whose height function f is Morse, all of whose critical points have index m (the middle dimension). For them K is of type F_{m-1} but not of type FP_m. Here B^m is complex hyperbolic space, and $\partial_\infty G = S^{2m-1}$.

The simplest is the *Cartwright–Steger surface* [3], with $m = 2$: a compact complex hyperbolic surface whose first Betti number is 2. Its map to the circle comes from its Albanese map to an elliptic curve, a Lefschetz fibration with exactly three nodes whose fibers have genus 19 (see Szabó [7] for the Albanese map and the fundamental group), and f has 3 critical points in X, all of index 2. So K is finitely generated but not finitely presented, and the zippers are path-connected subsets of S^3 (Theorem 1.1).

- `zippers draw cs` draws them. The default presentation has 36 letters and an element t that is a word of length 17 in the three generators of the lattice.
- For instance, letter 1 is $a_1^{-1}a_3^{-1}$, and its roof is $t^{-1}s_1t = s_{32}\,s_8\,s_{19}$. The 36 roofs have 6 letters on average, and at most 14.

5.3 A near example: a cusped hyperbolic 5-manifold, zippers on S^4

Italiano, Martelli and Migliorini [5] found a hyperbolic 5-manifold that fibers over the circle. It has finite volume but is not closed: it has cusps, so π_1 is relatively hyperbolic and Z^+ and Z^- are not disjoint; see Remark 1.3. The programs draw its arcs by the same construction (`zippers draw imm`). Near a cusp, an arc runs into the cusp's point at infinity, and the programs draw it through those points. The pictures show a close family resemblance to the zippers of the figure-eight knot complement, especially near the cusps.

6 Linking at infinity and homology in the middle dimension

Let $X = G\backslash B^m$ be one of the examples of Llosa Isenrich and Py. Every critical point p of the height f on B^m (there are finitely many up to G) has:

- an *ascending thimble* T_p^+, the points that flow up from p along the gradient of f; it is an m-dimensional plane, since the index of p is m;
- a *descending thimble* T_p^-, the points that flow down from p; also an m-plane.

Theorem 6.1 (work in progress)**.** *The closure of T_p^+ in the closed ball $B^m \cup S^{2m-1}$ meets the sphere at infinity in an $(m-1)$-sphere $v_p^+ \subset Z^+$; likewise T_p^- gives an $(m-1)$-sphere $v_p^- \subset Z^-$. Their linking numbers in S^{2m-1} are*

$$\mathrm{lk}(v_p^+, v_q^-) = \delta_{pq}$$

(with suitable orientations). Consequently $H_{m-1}(Z^\pm) \neq 0$, and with Theorem 4.1, $Z^\pm$ is $(m-2)$-connected but not $(m-1)$-connected.

So the connectivity of the zippers records exactly how finite K is: of type F_{m-1} (Theorem 4.1 gives $(m-2)$-connected zippers), but not FP_m (the linked spheres).

Sketch of a proof. Again a sketch only.

1. The gradient lines in T_p^+ that start at p are staircases: along them f increases linearly, and they are quasigeodesic, with constants that do not depend on p. So their endpoints lie in Z^+, and they make up an $(m-1)$-sphere at infinity.

2. T_p^+ and T_q^- are disjoint when $p \neq q$, and when $p = q$ they meet only at p, transversally. So the intersection number of the two disks in the $2m$-ball is δ_{pq}, and the linking number of their boundaries is the same.

3. A sphere in Z^+ that links a sphere of Z^- cannot be filled in the complement of the second, which contains Z^+. So it is essential in Z^+.

What the programs show. For the Cartwright–Steger surface ($m = 2$) the theorem predicts linked loops. The thimble loops $v_p^\pm$ need not lie in the drawn zippers, but the program `link3` finds loops in the drawn Z^+ and Z^-, made from relations of K, and computes their linking numbers: the 13 pairs of `data/s3/linked_pairs.txt` have linking number ± 1. One of them (pair 5, linking number -1) is the S^3 picture of the gallery: `zippers draw cs --pairs linked_pairs.txt --pairindex 5 --context`.

References

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